

Assessment Task No. 1 (CSST 102)

Part A

1. Product Recommendation
2. Recommend products that a customer is likely to want to buy, based on their shopping behavior.
3. Products viewed, items purchased before, items added to cart, search keywords, rating given.
4. A list of recommended products for that specific customer.
5. Customers with similar browsing / purchase habits tend to like similar products often bought together tend to be recommended together.
6. Customer purchase history, browsing history, product categories, ratings / reviews, and cart data from many users.

Part B

Traditional Programming

A programmer would need to manually write fixed rules, such as “if a customer buys a phone, recommend a phone case” or “if a customer views category X three times, show more items from category X”. These rules would need to be written and updated by hand for every possible situation, which becomes impossible to maintain as the number of products and customer behaviors grows.

Machine Learning

Instead of hand-coding every rule, the ML algorithm is given many examples of past customer data (inputs) paired with what those customers actually bought or liked (outputs). By analyzing these examples, the algorithm learns the underlying pattern connecting customer behavior to purchase decisions on its own, and can then use that learned relationship to recommend products to new customers.

Part C

Input	Output
1	2
2	4
3	6
4	8
5	?

Part D

A Machine Learning algorithm learns by being given many examples of inputs paired with their correct outputs, which together make up the training data. During training, the algorithm makes an initial guess at the relationship between the inputs and outputs, then compares its guesses to the actual outputs and adjusts itself to reduce the errors. This process repeats many times until the algorithm has learned an accurate relationship between input and output. Once trained, the algorithm can take a new, unseen input and use the learned relationship to make a prediction.

Reflection

Learning from data is useful because many real-world problems, like recommending products or translating languages, involve many patterns that are too complex and numerous for a programmer to write out as fixed rules. Manually coding every possible customer behavior or word combination would be time-consuming, incomplete, and hard to maintain as new data and situations arise. By learning directly from data, an ML algorithm can automatically discover these complex patterns and adapt to new examples without needing a programmer to anticipate every case in advance.